Sequences, Limits, Derivatives, and Riemann Integration

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A limit identifies which changes eventually become negligible; a derivative selects the best linear approximation; an integral accumulates such changes. This development follows Rudin's *Principles of Mathematical Analysis* and MIT 18.100B real-analysis notes, with each conclusion tied to its hypotheses.[1]

§1.1 A limit is an eventually quantified statement

For a sequence $a : \mathbb{N} \to \mathbb{R}$, write $a_n \to \ell$ when

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall n \geq N : \quad |a_n - \ell| < \varepsilon.$$

The cutoff N depends on the requested error but must work for every later index. For $a_n = n/(n+1)$, we have $|a_n - 1| = 1/(n+1)$; choosing $N + 1 > 1/\varepsilon$ establishes convergence.

By contrast, $(-1)^n$ cannot converge. Both its even subsequence, constantly 1, and its odd subsequence, constantly -1, would have to approach the same limit. Uniqueness of limits contradicts this.

§1.2 Limits ignore the value at the point

Let a be an accumulation point of the domain A. The statement $\lim_{x \to a} f(x) = L$ means

$$\forall \varepsilon > 0 \ \exists \delta > 0 : \quad x \in A, \quad 0 < |x - a| < \delta \Longrightarrow |f(x) - L| < \varepsilon.$$

The puncture is essential. If $f(x) = x^2$ for $x \neq 2$, but $f(2) = 17$, then $\lim_{x \to 2} f(x) = 4$, even though $f(2) = 17$.

Theorem 1.2.1 (Sequential criterion)
The limit $f(x) \to L$ as $x \to a$ exists if and only if every sequence $x_n \in A \setminus \{a\}$ with $x_n \to a$ satisfies $f(x_n) \to L$.

Proof. The forward direction follows by applying the δ condition once x_n is close to a. Conversely, if no suitable δ exists for some $\varepsilon_0 > 0$, choose for each n a point $x_n \in A$ such that

$$0 < |x_n - a| < 1/n, \qquad |f(x_n) - L| \geq \varepsilon_0.$$

Then $x_n \to a$, but its image does not converge to L, a contradiction. □

This proves discontinuity of the rational-indicator function at every real point: rational and irrational sequences converge to any chosen point but have constantly different function values.

[1] https://ocw.mit.edu/courses/18-100b-real-analysis-spring-2025/pages/lecture-notes/

§1.3 The derivative is a linear approximation

A function f is differentiable at a if

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists. Equivalently,

$$f(a+h) = f(a) + f'(a)h + o(|h|),$$

where $r(h) = o(|h|)$ means $r(h)/|h| \to 0$. The linear part $f'(a)h$ is unique, because two proposed coefficients would differ by $o(|h|)/h \to 0$.

For $f(x) = x^2$,

$$f(a+h) = a^2 + 2ah + h^2 = a^2 + 2ah + o(|h|).$$

The little-o term measures the actual approximation error. By contrast $r(h) = O(h^2)$ asserts only that $|r(h)| \le Ch^2$ near zero, a stronger scale-specific bound in this example.

The product rule follows from the exact identity

$$u(a+h)v(a+h) - u(a)v(a) = u(a+h)(v(a+h) - v(a)) + v(a)(u(a+h) - u(a)).$$

Divide by h and use differentiability of both factors to obtain

$$(uv)' = u'v + uv'.$$

Applying this to $v(1/v) = 1$ gives the quotient rule when $v \neq 0$. Composing two first-order expansions gives the chain rule $(F \circ g)' = F'(g)g'$. Repeated product differentiation leads by induction to

$$(uv)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} u^{(k)} v^{(n-k)}.$$

§1.4 An integral is a partition-independent sum

For $P : a = x_0 < \cdots < x_m = b$, select $\xi_i \in [x_{i-1}, x_i]$ and define

$$S(f, P, \xi) = \sum_{i=1}^{m} f(\xi_i)(x_i - x_{i-1}).$$

Riemann integrability requires these sums to approach one limit as the maximum interval length tends to zero, independently of the sample points. Every continuous function on a compact interval is integrable.

For $f(x) = x$ on $[0, 1]$, equal partitions and right-endpoint samples give

$$S_n = \frac{1}{n^2} \sum_{i=1}^{n} i = \frac{n(n+1)}{2n^2} \to \frac{1}{2}.$$

This computes the value once integrability has been established by continuity.

§1.5 The fundamental theorem connects both operations

Theorem 1.5.1 (Fundamental theorem of calculus)
For continuous $f : [a, b] \to \mathbb{R}$, the function

$$F(x) = \int_a^x f(t)\,dt$$

has derivative $F'(x) = f(x)$ at interior points. If $G' = f$, then $\int_a^b f(t)\,dt = G(b) - G(a)$.

For $h > 0$,

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,dt.$$

The difference from $f(x)$ has absolute value at most $\sup_{|t-x|\le h} |f(t) - f(x)|$, which tends to zero by continuity. The case $h < 0$ is analogous. The second identity follows from the first because $F - G$ has derivative zero on the connected interval, and the mean value theorem makes it constant.

§1.6 What changes when one changes the norm

Metric convergence replaces $|a_n - a|$ by $d(a_n, a)$. A useful example is $f_n(x) = x^n$ on $[0, 1]$. Although

$$\|f_n\|_{L^1} = \int_0^1 x^n\,dx = \frac{1}{n+1} \to 0,$$

its supremum norm is $\|f_n\|_\infty = 1$ for every n. Thus it converges to zero in L^1 but not uniformly. Different notions of distance ask for different notions of small error, even though every convergence proof follows the same epsilon-quantifier pattern.